

Two ladders, and one rule that decides most of the marks: **read the unit on every number before you touch it.** Working, always.

1. Draw one line from each quantity on the left to the equal quantity on the right. Not all will be used. [5]

1 byte	8 000 bits
1 kilobyte	1 024 gibibytes
1 kibibyte	8 bits
1 megabyte	1 024 kilobytes
1 tebibyte	1 000 kilobytes
	8 192 bits

2. Prefixes. [5]

(a) Complete the description. [4]

A mebibyte has a _____ prefix. Four mebibytes is the same as _____ kibibytes.

A gigabyte has a _____ prefix. Three terabytes is the same as _____ gigabytes.

(b) Tick (✓) **one** box only to identify the **largest** file size. [1]

☐ 2000 kibibytes ☐ 2 mebibytes ☐ 0.2 megabytes ☐ 2000 kilobytes

3. A bitmap image. [4]

A scanner produces a bitmap image with a resolution of **1024 × 768** pixels and a colour depth of **24 bits** per pixel.

(a) Estimate the file size in **mebibytes**. Show your working. [3]

(b) The question says *estimate*, not *calculate*. State **one** reason the real file will be larger than your answer. [1]

4. A sound file. [4]

A **32 second** clip is recorded with a sampling rate of **16 384 Hz** and a sampling resolution of **16 bits**.

(a) Calculate the file size in **kibibytes**. Show your working. [3]

(b) The clip is re-recorded with the sampling rate **halved**. Give the new file size. [1]

5. Find the mistake. A student calculates the size of an 800×600 image with a colour depth of 8 bits, giving the answer in KiB. [3]

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800 x 600      = 480 000
480 000 x 8     = 3 840 000 bits
3 840 000 ÷ 1024 = 3750 KiB
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Identify the mistake, and give the correct answer.

6. State the difference between a gibibyte and a gigabyte. [1]

🍌 **The ladder this week — Karel, not Python.** No ice-cream challenge for this lesson by design: the code you write this week is **Karel 07 PotholePatrol** and **08 HurdleJumper**, your first conditions. The two calculators in the deck are the other half — go back and change a number in them.

— **Want to dig?** Your syllabus stops the binary ladder at **tebi**, and the IGCSE one stops at **exbi**. The ladder itself does not stop there — **zebi** (2^{70}) and **yobi** (2^{80}) are standard units too, and almost nobody has ever needed them. Then in **2022** the body that governs the metric system did something it had not done in three decades: it added two brand-new prefixes at the top, **ronna** (10^{27}) and **quetta** (10^{30}). Find out roughly how much data the world produces in a year, work out which rung of the ladder that sits on, and you will see exactly why they were in a hurry.

Additional space. If you need more room for any answer, continue here and write the question number in the margin beside it.
